

AA 242 A

Coordinate Systems:

$$\vec{r} = r\hat{r} + z\hat{z}$$

$$\vec{v} = \dot{r}\hat{r} + r\dot{\hat{r}} + \dot{z}\hat{z} + z\dot{\hat{z}}$$

$$= \dot{r}\hat{r} + r\dot{\phi}\hat{\phi} + \dot{z}\hat{z}$$

Auxiliary: $\dot{\hat{r}} = \vec{\omega} \times \hat{r} = \dot{\phi}\hat{\phi}$

$$\dot{\hat{\phi}} = \vec{\omega} \times \hat{\phi} = -\dot{\phi}\hat{r}$$

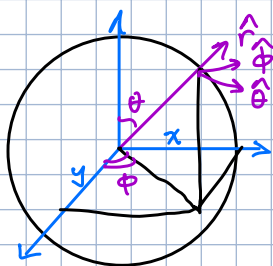
$$\vec{r} = r\hat{r} \quad \vec{v} = \dot{r}\hat{r} + r\dot{\hat{r}}$$

$$\dot{\hat{r}} = \vec{\omega} \times \hat{r}$$

$$= (\dot{\phi}\hat{\phi} + \dot{z}\hat{z}) \times \hat{r}$$

$$= \dot{\phi}\hat{\theta} + \dot{\phi}\sin\theta\hat{\phi}$$

$$\vec{v} = \dot{r}\hat{r} + r\dot{\phi}\hat{\theta} + r\dot{\phi}\sin\theta\hat{\phi}$$



Lagrange & D'Alembert's

$$\vec{F}_i + \vec{f}_i - m_i \ddot{\vec{r}}_i = 0$$

- $\vec{F}_i$ $\hat{=}$ Applied forces
- $\vec{f}_i$ $\hat{=}$ forces of constraint
- $\vec{f}_i$ $\hat{=}$ inertial forces

$$\delta W = \sum_{i=1}^N (\vec{F}_i + \vec{f}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0$$

if $\vec{f}_i$ is WORKLESS CONSTR.

$$\delta W = \sum_{i=1}^N (\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0$$

$$\delta \vec{r}_i = \sum_{j=1}^{3N-M} \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j \hat{=} Q_j$$

$$\Rightarrow \sum_{i=1}^N \vec{F}_i \cdot \delta \vec{r}_i = \sum_{i,j} \vec{F}_i \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j$$

Q_j : component of GENERALIZED FORCE,

Workless Constraint:

$$\sum_{i=1}^N \vec{f}_i \cdot \delta \vec{r}_i = 0 \quad \vec{f}_i \hat{=} \text{constraint force}$$

$$\delta \vec{r}_i \hat{=} \text{virtual displacement}$$

Rigid Rods, Frictionless Slide, Rolling

BASIC MATH

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\text{proj}_{\vec{a}} \vec{b} = \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$$

2nd-Order ODEs. $ay'' + by' + c = 0$

r_1/r_2 : $y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

r_1/r_1 : $y(t) = C_1 e^{r_1 t} + t C_2 e^{r_1 t}$

$\lambda \pm Mi$: $y(t) = C_1 e^{\lambda t} \cos \mu t + C_2 e^{\lambda t} \sin \mu t$

Rotating FOR

Quick Derivation of the Golden Rule.

$$\dot{\vec{r}} = \dot{r}\hat{r} + r\dot{\hat{r}} = \dot{r}\hat{r} + r(\vec{\omega} \times \hat{r}) = \dot{r}\hat{r} + \vec{\omega} \times \vec{r}$$

$$\dot{\vec{r}}_S = \dot{\vec{r}}_B + \vec{\omega}_{B/S} \times \vec{r}$$

Coriolis, Centrifugal, & Euler

$$\frac{d\vec{r}}{dt}|_S = \vec{v}_S = \vec{v}_R + \vec{\omega} \times \vec{r} \quad // \text{Apply Golden Rule 1}$$

$$\frac{d\vec{v}_S}{dt}|_S = \vec{a}_S = \frac{d\vec{v}_S}{dt}|_R + \vec{\omega} \times \vec{v}_S \quad // \text{App. Golden Rule 2}$$

$$= \frac{d}{dt}(\vec{v}_R + \vec{\omega} \times \vec{r})|_R + \vec{\omega} \times (\vec{v}_R + \vec{\omega} \times \vec{r})$$

$$= \frac{d\vec{v}_R}{dt} + \frac{d\vec{\omega}}{dt}|_R \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}|_R + \vec{\omega} \times \vec{v}_R + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\vec{a}_S = \frac{d\vec{v}_R}{dt} + (\vec{\alpha} \times \vec{r}) + 2(\vec{\omega} \times \vec{v}_R) + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\vec{a}_S = \vec{a}_R + (\vec{\alpha} \times \vec{r}) + 2(\vec{\omega} \times \vec{v}_R) + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\Rightarrow \vec{m}\vec{a}_R = \vec{m}\vec{a}_S - \underbrace{(\vec{\alpha} \times \vec{r})}_{\text{EULER}} - \underbrace{2(\vec{\omega} \times \vec{v}_R)}_{\text{CORIOLIS}} - \underbrace{\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{CENTRIFUGAL}}$$

Conservation

Linear Momentum: $\vec{p} = m \frac{d\vec{R}}{dt} = \sum_i m_i \vec{v}_i$

Energy: $KE = \frac{1}{2} m v^2 = \frac{1}{2} m \vec{v} \cdot \vec{v} = \frac{1}{2} I \omega^2 = \frac{1}{2} I (\dot{\phi})^2$

$V = mgh$ or $\frac{1}{2} k x^2$

Ang. Momentum: $\vec{H} = \vec{r} \times \vec{p} = \sum_i \vec{r}_i \times m_i \vec{v}_i$

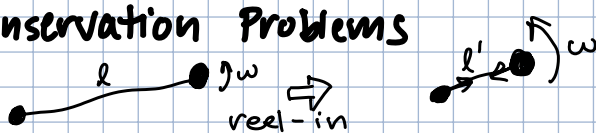
$$= mvr = m\omega r^2$$

Rocket Equ.

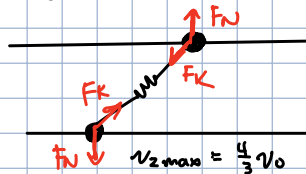
$$(m - dm)(v + dv) + dm(v - u) - m_0 v_0 = \Delta \vec{p}$$

Thrust: $-u \frac{dm}{dt}$ max velocity limited by $\frac{u_0}{m_0}$

Conservation Problems



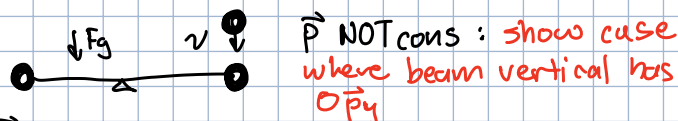
Energy NOT conserved $\rightarrow$ Someone applied work
 Linear Momentum conserved $\rightarrow \Sigma F = 0!$
 Angular Momentum conserved $\rightarrow \Sigma M = 0!$



Energy is conserved,
 $\frac{dE}{dt} = 0$ & rails are frictionless

$\vec{P}_x$ cons. - $\vec{F}$ forces are internal

$\vec{H}$ NOT cons $\vec{F}_N$ forces not radial



$\vec{P}$ NOT cons: show case where beam vertical has $0 \vec{P}_y$

$\vec{H}$ conserved AT COLLISION

$\vec{H}$ not cons. AFTER COLLISION (gravity now acts as imbalanced torque)

E is NOT cons. $\rightarrow$ inelastic 'sticky' collision

E cons. AFTER COLLISION

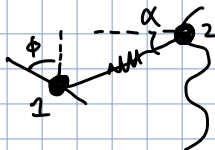


E is conserved elastic collision

$\vec{p}$ is cons. no $\vec{F}_{ext}$

$\vec{H}$ is cons no $\vec{M}_{ext}$.

Rod CM will only translate in x-dir!



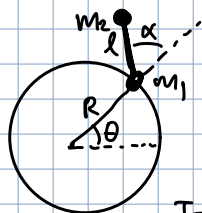
Gen. Coords: x_1, y_1, x_2, y_2

Constraints: $x_2 = \sin(\alpha y_2)$

$f_1 = x_2 - \sin(\alpha y_2) = 0$

$\dot{x}_1 \cos \phi = -\dot{y}_1 \sin \phi$

$f_2 = \dot{x}_1 \cos \phi + \dot{y}_1 \sin \phi = 0$



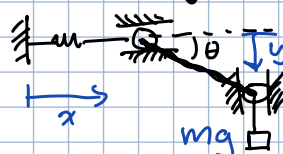
$\vec{v}_1 = R\dot{\theta}\hat{\theta}$

$\vec{v}_2 = \vec{v}_1 + l(\dot{\alpha} + \dot{\theta})(-\sin \alpha \hat{r} + \cos \alpha \hat{\theta})$

$= -l(\dot{\theta} + \dot{\alpha}) \sin \alpha \hat{r} + (R\dot{\theta} + l(\dot{\theta} + \dot{\alpha}) \cos \alpha) \hat{\theta}$

$T = \frac{1}{2} m (\vec{v}_1 \cdot \vec{v}_1 + \vec{v}_2 \cdot \vec{v}_2)$

Lagrangian Problems.



gen. coords: x, y, θ

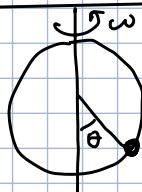
constraints:

$y = l \sin \theta \quad f_1 = y - l \sin \theta$

$x + l \cos \theta = C \quad f_2 = x - l \cos \theta$

$T = \frac{1}{2} m \dot{y}^2$

$V = \frac{1}{2} k x^2 - mgy$



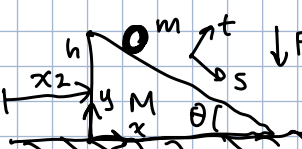
Gen. Coords: r, θ, ϕ

constraints: $r = \text{const.}$

$f_1 = r - R = 0$

$\omega = \text{const} = \dot{\phi} \quad f_2 = \phi - \omega t = 0$

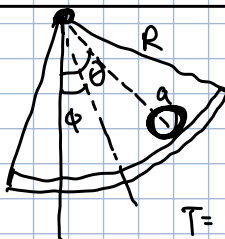
$T = \frac{1}{2} m (a^2 \dot{\theta}^2 + a^2 \omega^2 \sin^2 \theta) \quad V = mga(1 - \cos \theta)$



Gen Coords:

x, y, x_2

Constraints: $t = \text{const.} \quad f_1 = t - C = 0$



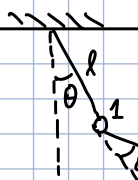
Gen Coords: θ, ϕ

constraint: (semiholonomic)

$\dot{\theta} = -\frac{a\omega}{R} + \dot{\phi} \quad f_1 = \dot{\theta} - \dot{\phi} + \frac{a\omega}{R}$

$T = \frac{1}{2} m (R\dot{\phi})^2 + \frac{1}{2} m (R\dot{\theta})^2 + \frac{1}{2} I_{\text{hoop}} \omega^2$

$V = mgR(1 - \cos \theta) + mgR(1 - \cos \phi)$



$\vec{r}_1 = l \sin \theta \hat{x} - l \cos \theta \hat{y}$

$\dot{\vec{r}}_1 = l \dot{\theta} \cos \theta \hat{x} + l \dot{\theta} \sin \theta \hat{y}$

$\vec{r}_2 = \vec{r}_1 + l \sin(\theta + \phi) \hat{x} - l \cos(\theta + \phi) \hat{y}$

$\dot{\vec{r}}_2 = \dot{\vec{r}}_1 + l(\dot{\theta} + \dot{\phi}) \cos(\theta + \phi) \hat{x}$

$+ l(\dot{\theta} + \dot{\phi}) \sin(\theta + \phi) \hat{y}$

$T = \frac{1}{2} m (\dot{\vec{r}}_1^2 + \dot{\vec{r}}_2^2)$

Allowed on
MT

AA 242A

TOPIC 10: Lagrange's Equations

Equation	Constraint	Conservative forces only?	Forces only derivable from scalar potential?	DOF problem?
$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j$ where Q_j are all generalized forces	Holonomic constraints, where the constraint equations are used to eliminate redundant coordinates	No	No	Use if $\# q_j = \text{DOF}$
$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial (T - V)}{\partial q_j} = 0$		No	Yes	
$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0$		Yes*	Yes	
$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = Q_j'$ where Q_j' are generalized forces not accounted for in V		No	No	
$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} + \sum_{k=1}^m \lambda_k \frac{\partial f_k}{\partial q_j} = 0$	m holonomic constraints f_k	Yes	Yes	Use if $\# q_j > \text{DOF}^{**}$
$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} + \sum_{k=1}^m \lambda_k \frac{\partial f_k}{\partial q_j} = Q_j'$ where Q_j' are generalized forces not accounted for in V and λ		No	No	
$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} + \sum_{k=1}^m \mu_k \frac{\partial f_k}{\partial \dot{q}_j} = 0$	m semi-holonomic where the constraints f_k are linear in rates	Yes	Yes	Use if $\# q_j \geq \text{DOF}^{**}$
$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} + \sum_{k=1}^m \mu_k \frac{\partial f_k}{\partial \dot{q}_j} = Q_j'$ where Q_j' are generalized forces not accounted for in V and μ		No	No	

*The definition of "time" is ambiguous, so while technically these potentials can depend on time (thereby invalidating the definition of a conservative force), this equation is considered the conservative form of Lagrange's equation.

**Can also use if it's inconvenient to reduce all generalized coordinates to independent coordinates, or to get the forces of constraint.